Cheat sheet: regulated diagnostic biostatistics

[S] = verified in a primary public source (2026-10-16); **[Y]** = synthesis. "NV" = not verified: paywalled.

Document map (CLSI editions verified on clsi.org)

Question	Document
Precision	**EP05 Ed4 (2025-12)**; A3 (2014): 20 days × 2 runs × 2 reps
Verify precision/bias	EP15-A3 (2014): ≤5 working days; bias ≥25 results over ≥5 days
LoB/LoD/LoQ	EP17-A2 (2012)
Linearity	EP06 Ed2 (2020)
Method comparison	EP09c Ed3 (2018)
Interference	EP07 Ed3 (2018); EP37 tables
Stability / lots	**EP25 Ed2 (2023)**; **EP26 Ed2 (2022)**
Binary tests	EP12 Ed3 (2023)
ROC	EP24-A2 (2011)
Reference intervals	EP28-A3c (2010)
Framework	**EP19 Ed3 (2022)**; specimen types **EP35 Ed2 (2025)**

Design counts for most EP documents: NV.

Formulas

Topic	Formula
Within-lab SD	$\sqrt{(\sigma^2 day + \sigma^2 run + \sigma^2 rep)}$; CV% = 100·SD/mean [Y]
Satterthwaite df	$(\Sigma c_i MS_i)^2 / \Sigma (c_i MS_i)^2/df_i$; 20×2×2 ≈ 70 df
UVL	$claim \cdot \sqrt{(\chi^2_{0.95}, \nu/\nu)}$ [Y]
LoB / LoD	mean_b + 1.645 SD_b; LoB + 1.645 SD_low
Probit LoD95	$10^{((1.645 - b_0)/b_1)}$
Bias at Xc	a + (b − 1)·Xc
Total error	|bias| + 1.96 SD
PPV	Se·p / (Se·p + (1−Sp)(1−p))
LR+ / LR−	Se/(1−Sp); (1−Se)/Sp
Wilson CI	$(\hat{p} + z^2/2n \pm z\sqrt{(\hat{p}\hat{q}/n + z^2/4n^2)}) / (1 + z^2/n)$
n cases (Se ± d)	$z^2 p(1-p)/d^2$; total = cases/prevalence
AUC	P(case > control); DeLong var from structural components
RCV	$\exp(\pm 1.96 \sqrt{2}\ SD_w) - 1$
Deming slope	$(s_yy - \delta s_xx + \sqrt{((s_yy - \delta s_xx)^2 + 4\delta s_xy^2)}) / 2s_xy$

Regulatory anchors

Item	Fact
IVDR Art 47 / 48	classification / conformity assessment
Art 56 / 57 / 58	performance evaluation / study requirements / additional (58(2) CDx)
Annex I 9.1(a)(b)	analytical / clinical parameters
Annex VIII	**Rule 3(f): CDx = class C**
Art 48(7)(8)	CDx: consult competent authority (2001/83/EC) or EMA
Annex XIII	Part A PEP, PER, CPSP 2.3.2; Part B PMPF
Art 2(7)(38)(40)(41)	CDx; scientific validity of analyte; analytical; clinical performance
Transition 2024/1860	legacy to 31 Dec 2027; D 2027, C 2028, B 2029
MDCG 2022-2	clinical evidence for IVDs (Jan 2022)
ISO 20916:2019	clinical performance studies, good study practice
510(k) / De Novo / PMA	cleared / granted / approved [Y]
FDA 2007	PPA/NPA, not Se/Sp, vs non-reference
FDA CDx	2014 final; 2016 codevelopment **draft**; 2020 "Specific Group of Oncology…" final
STARD / QUADAS-2	30 items / 4 domains
ICH E9, E9(R1)	1998, 2019 (by analogy)
CDISC	SDTMIG-MD v1.1, ADaMIG-MD v1.0; no IVD standard found

Elecsys pTau217 (FDA K261686, decision 2026-08-19)

Item	Fact
Cut-offs pg/mL	neg ≤ 0.223; int > 0.223–≤ 0.378; pos > 0.378
Pivotal	N 958; PET+ 378 (39.5%); 6-reader visual read
3×2 (PET+/PET−)	pos 285/37; int 72/119; neg 21/424
Headline	PV pos 88.5%; NPV 95.3%; intermediate 19.9%; PPA 75.4%; NPA 73.1%
Not in summary	eGFR analysis; monitoring exclusion (Roche release only)

Item	Fact
Papers	prototype assay; Hibar 0.189 single cut-off; Hortsch 5.78% indeterminate

Rules of thumb [Y]

- Se/Sp need a reference standard; otherwise PPA/NPA.
- Lock cut-offs before validation; fix the ≤ / < convention.
- Intermediates: report fraction, per-category PV, PPA/NPA with intermediates as misses.
- Missing bridging samples: ask why; MI needs arm × outcome term; tipping-point.
- Non-significant ≠ equivalent: use CI inside the limit.
- Mark NV items as unknown in the interview.